FAULT CURRENT STUDIO - ENGINEERING REPORT
BROWSER EXPORT - ENGINEERING REVIEW REQUIRED

===== SUMMARY =====
Project: Fault Current Studio advanced study
Method: protectionrange
Model status: ENTERED_ENVELOPE_MARGIN_MET_ALL_LOCATIONS
Captured UTC: 2026-10-26T06:04:35.813Z
Warning: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Report data provenance: Entered study inputs, isolated calculated method outputs. No external OEM certification or auto-selected breaker/PPE.

===== STUDY INPUTS =====
Input path: requiredMarginS | Entered value: 0.15
Input path: downstream.model | Entered value: EXAMPLE_DOWNSTREAM_125A
Input path: downstream.source | Entered value: ILLUSTRATIVE POINTS NOT OEM DATA: replace with publication and actual settings
Input path: downstream.revision | Entered value: illustrative-1
Input path: downstream.curveCsv | Entered value: 100,0.65,0.8?200,0.25,0.3?400,0.08,0.10?800,0.03,0.04
Input path: upstream.model | Entered value: EXAMPLE_UPSTREAM_400A
Input path: upstream.source | Entered value: ILLUSTRATIVE POINTS NOT OEM DATA: replace with publication and actual settings
Input path: upstream.revision | Entered value: illustrative-1
Input path: upstream.curveCsv | Entered value: 100,1.4,1.7?200,1.0,1.2?400,0.7,0.9?800,0.38,0.50
Input path: faultRanges[0].location | Entered value: Illustrative downstream feeder faults
Input path: faultRanges[0].source | Entered value: ILLUSTRATIVE electrical fault study, replace with signed fault cases
Input path: faultRanges[0].downstreamMinA | Entered value: 130
Input path: faultRanges[0].downstreamMaxA | Entered value: 700
Input path: faultRanges[0].upstreamPerDownstream | Entered value: 1
Input path: faultRanges[1].location | Entered value: Illustrative two-source scenario
Input path: faultRanges[1].source | Entered value: ILLUSTRATIVE current distribution, replace with signed case
Input path: faultRanges[1].downstreamMinA | Entered value: 230
Input path: faultRanges[1].downstreamMaxA | Entered value: 600
Input path: faultRanges[1].upstreamPerDownstream | Entered value: 0.8

===== EQUIPMENT DUTY =====
Calculated output path: status | Value: ENTERED_ENVELOPE_MARGIN_MET_ALL_LOCATIONS
Calculated output path: worstCaseMarginS | Value: 0.37973736355067933
Calculated output path: requiredMarginS | Value: 0.15
Calculated output path: worstLocation | Value: Illustrative downstream feeder faults
Calculated output path: worstFaultPrimaryA.downstream | Value: 700
Calculated output path: worstFaultPrimaryA.upstream | Value: 700
Calculated output path: scenarios[0].location | Value: Illustrative downstream feeder faults
Calculated output path: scenarios[0].source | Value: ILLUSTRATIVE electrical fault study, replace with signed fault cases
Calculated output path: scenarios[0].downstreamRangeA.minimum | Value: 130
Calculated output path: scenarios[0].downstreamRangeA.maximum | Value: 700
Calculated output path: scenarios[0].upstreamPerDownstream | Value: 1
Calculated output path: scenarios[0].worstCase.downstreamPrimaryA | Value: 700
Calculated output path: scenarios[0].worstCase.upstreamPrimaryA | Value: 700
Calculated output path: scenarios[0].worstCase.downstreamMaximumClearingS | Value: 0.04772227898715389
Calculated output path: scenarios[0].worstCase.upstreamMinimumClearingS | Value: 0.4274596425378332
Calculated output path: scenarios[0].worstCase.marginS | Value: 0.37973736355067933

Calculated output path: scenarios[0].candidateCount | Value: 5
Calculated output path: scenarios[1].location | Value: Illustrative two-source scenario
Calculated output path: scenarios[1].source | Value: ILLUSTRATIVE current distribution, replace with signed case
Calculated output path: scenarios[1].downstreamRangeA.minimum | Value: 230
Calculated output path: scenarios[1].downstreamRangeA.maximum | Value: 600
Calculated output path: scenarios[1].upstreamPerDownstream | Value: 0.8
Calculated output path: scenarios[1].worstCase.downstreamPrimaryA | Value: 600
Calculated output path: scenarios[1].worstCase.upstreamPrimaryA | Value: 480
Calculated output path: scenarios[1].worstCase.downstreamMaximumClearingS | Value: 0.058508641898213344
Calculated output path: scenarios[1].worstCase.upstreamMinimumClearingS | Value: 0.5960891447185928
Calculated output path: scenarios[1].worstCase.marginS | Value: 0.5375805028203794
Calculated output path: scenarios[1].candidateCount | Value: 5
Calculated output path: candidateCount | Value: 10
Calculated output path: downstream.model | Value: EXAMPLE_DOWNSTREAM_125A
Calculated output path: downstream.source | Value: ILLUSTRATIVE POINTS NOT OEM DATA: replace with publication and actual settings
Calculated output path: downstream.revision | Value: illustrative-1
Calculated output path: upstream.model | Value: EXAMPLE_UPSTREAM_400A
Calculated output path: upstream.source | Value: ILLUSTRATIVE POINTS NOT OEM DATA: replace with publication and actual settings
Calculated output path: upstream.revision | Value: illustrative-1
Calculated output path: approved | Value: false
Calculated output path: method | Value: Exact minimum of manufacturer ENTERED monotone log-log time envelope on continuous source-current interval via knots and analytic stationary points
Calculated output path: limitations[0] | Value: User-digitised OEM source points are not independently authenticated here
Calculated output path: limitations[1] | Value: Instantaneous and energy-limiting selectivity must use official manufacturer tables/testing where applicable
Calculated output path: limitations[2] | Value: Upstream-to-downstream current ratio is a documented constant per entered location range; verify all independent sources and CTs
Calculated output path: limitations[3] | Value: No CT transient saturation, reset, zone selectivity, protective settings verification, or OEM approval

===== ASSUMPTIONS & QA =====
Evidence / limitation: Release status | Value: BOUNDED METHOD, NOT A STANDARDS/DEVICE APPROVAL
Evidence / limitation: Warning | Value: ENGINEERING SCREENING ONLY - independently verify actual manufacturer, utility, installation, equipment settings and standards edition before professional use.
Evidence / limitation: Study method | Value: protectionrange
Evidence / limitation: Timestamp | Value: 2026-10-26T06:04:35.813Z
Evidence / limitation: outputs.limitations[0] | Value: User-digitised OEM source points are not independently authenticated here
Evidence / limitation: outputs.limitations[1] | Value: Instantaneous and energy-limiting selectivity must use official manufacturer tables/testing where applicable
Evidence / limitation: outputs.limitations[2] | Value: Upstream-to-downstream current ratio is a documented constant per entered location range; verify all independent sources and CTs
Evidence / limitation: outputs.limitations[3] | Value: No CT transient saturation, reset, zone selectivity, protective settings verification, or OEM approval